

RESULTS

Results

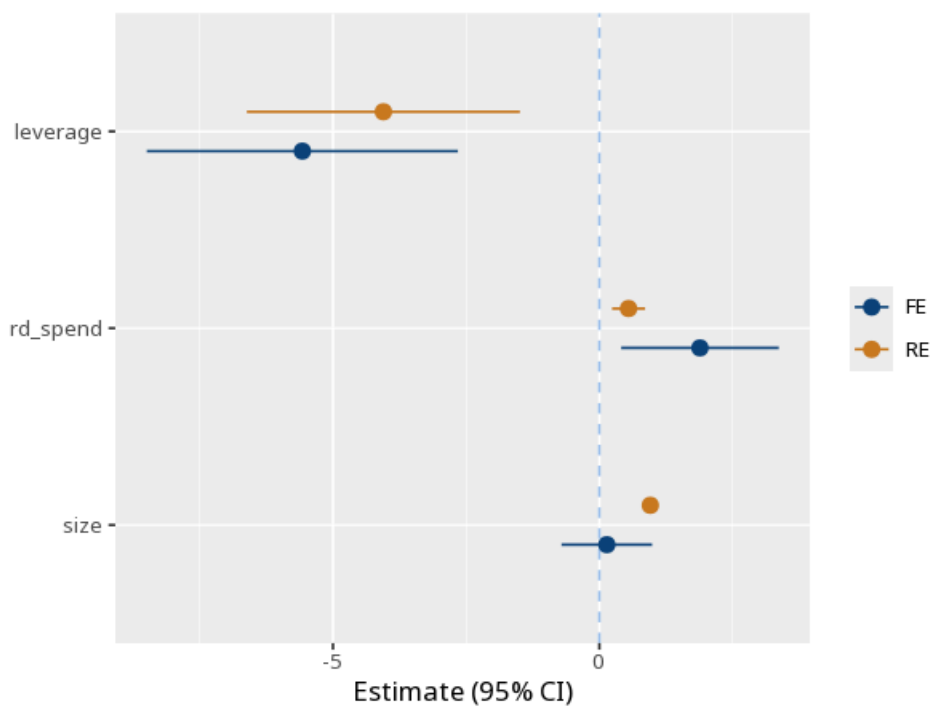
01 · HAUSMAN TEST

fixed vs. random effects?

Table 1. Hausman test

	Fixed effects	Random effects	Difference
leverage	−5.57 (1.47) [−8.49, −2.66]	−4.05 (1.29) [−6.62, −1.49]	−1.52
rd_spend	1.89 (0.74) [0.41, 3.37]	0.55 (0.16) [0.24, 0.86]	1.34
size	0.14 (0.43) [−0.71, 0.99]	0.96 (0.05) [0.86, 1.05]	−0.82
Num.Obs.	96		
N entities	12		
R ²	.91	.98	
Hausman $\chi^2(3) = 3.07$, p .381			

Figure. FE vs. RE



Understanding the numbers

Fixed effects. The fixed-effects estimate for this predictor - consistent even if the random-effects assumption fails. Here, the fixed-effects estimate is -5.57 .

Random effects. The random-effects estimate for the same predictor - more efficient, but only valid if entity effects are uncorrelated with the predictors. Here, the random-effects estimate is -4.05 .

Difference. The difference between the fixed- and random-effects estimates - the quantity the Hausman test evaluates. Here, the difference is -1.52 .

N. The total number of entity-period observations underlying both models. Here, $N = 96$.

N entities. The number of distinct entities in the panel. Here, $N \text{ entities} = 12$.

R². Each model's own R², reported alongside the Hausman comparison. Here, R² is FE .91 / RE .98.

Hausman χ^2 . The Hausman test statistic - how far the fixed- and random-effects estimates diverge, in aggregate across predictors. Here, $\chi^2(3) = 3.07$, $p = .381$.

How to read this test

Tests whether the random-effects assumption holds. A p below alpha means the estimates differ systematically $\rightarrow$ use fixed effects; a non-significant p means random effects is acceptable (and more efficient). Method: R plm package (Croissant & Millo, 2008). The standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Your significance threshold (α) is 0.05.

APA template: A Hausman test comparing the fixed- and random-effects estimates gave $\chi^2(3)=3.07$, $p = .381$.

Statistical basis: Hausman specification test (Hausman, J. A. (1978). "Specification tests in econometrics." *Econometrica*, 46(6), 1251-1271.); Implementation (plm package) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." *Journal of Statistical Software*, 27(2), 1-43.). Full references in the exported CITATIONS.txt.

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